

Response Letter with Annotations

Bailey, E. R., Matz, S. C., Youyou, W., & Iyengar, S. S. (2020). Authentic self-expression on social media is associated with greater subjective well-being. *Nature Communications*, 11(1), 4889.

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Annotations

1. Purposes

- a. Motivation (one-sentence summary)
To test whether authentic, rather than self-idealized, self-expression on social media is associated with greater subjective well-being.
- b. Specific Rationale (several bullet points)
 - Prior authenticity research relies heavily on self-reports, which may be distorted by social desirability and current mood.
 - Social media creates a tension between presenting an idealized self and expressing one's actual personality.
 - The authors introduce a behavioral/computational measure "Quantified Authenticity" to compare self-reported personality with personality inferred from Facebook Likes and status updates.
 - They also address the causal-direction problem by following the large-scale correlational study with a preregistered longitudinal experiment.
- c. Research Questions and/or Hypotheses
 - Hypothesis: Greater authentic self-expression on social media will be positively associated with Life Satisfaction.
 - Exploratory question: Do people with more socially desirable personality profiles benefit more from authentic self-expression than others?
 - Identify IV, DV, mediator, moderator
 - Main IV: Quantified Authenticity (Study 1); authenticity vs. self-idealization posting condition (Study 2)
 - DVs: Life Satisfaction (Study 1 and Study 2), mood, positive affect, negative affect (Study 2)
 - Mediator: None formally tested
 - Moderator: Big Five personality traits / socially desirable personality profile (exploratory moderation test)
 - Identify specific relationships among variables
 - Greater authentic self-expression → higher Life Satisfaction, subjective well-being

2. Method

- a. Research Design
 - Study 1 is an observational, cross-sectional analysis of archival Facebook-based survey data linked to digital trace measures. Study 2 is

longitudinal and experimental because the same participants were measured repeatedly across two intervention periods.

- Study 2 used a preregistered 2-week crossover experiment. Participants were randomly assigned to complete two 7-day interventions in different orders: one week posting authentically and one week posting in a self-idealized way. At t0 they completed baseline measures and reviewed personality feedback; at t1 and t2 they completed the same well-being measures after each intervention week.

b. Sample (representative sample?)

- Study 1: 10,560 Facebook users from the myPersonality project who completed personality and Life Satisfaction measures; 9,237 qualified for Likes-based models and 3,215 for language-based models.
- Study 2: 90 students/social media users from Northeastern University (M age = 22.98; 72.22% female). The preregistered target was 200, but data collection stopped early because of COVID-19 lab closure.
- Representativeness: Neither sample is nationally representative; Study 1 is large and behavior-based but limited to Facebook users who volunteered data, and Study 2 is a convenience sample of university-affiliated participants.

c. Measures

- Self-reported personality: IPIP in Study 1; BFI-2S in Study 2 pre-screen.
- Computer-predicted personality: Facebook Likes (10-fold cross-validated LASSO; average $r = .56$ with self-reports) and status-update language (cross-validated Ridge regression; average $r = .41$).
- Quantified Authenticity: inverse Euclidean distance between self-reported and predicted Big Five personality profiles after z-standardization; higher scores indicate greater authenticity. Robustness checks used Manhattan distance, cosine similarity, and correlational similarity.
- Well-being: Satisfaction With Life Scale in Study 1 and Study 2; Study 2 also included positive affect, negative affect, and a single-item mood measure.
- Controls in Study 1: self-reported Big Five traits, personality extremeness; robustness checks also considered age and gender.

d. Analysis

- Study 1: OLS regression predicting Life Satisfaction from Quantified Authenticity, first without controls and then with personality controls; exploratory interaction models tested moderation by Big Five traits.
- Additional analyses separated normative self-enhancement from self-deprecation and tested alternative authenticity metrics.
- Study 2: independent-samples t tests for between-person comparisons and dependent-samples t tests for within-person comparisons across authentic vs. self-idealized weeks.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Hypothesis supported: In Study 1, greater authenticity was positively associated with Life Satisfaction for both Likes-based and language-based measures.

- The moderation expectation was not supported consistently: people with more socially desirable traits did not reliably benefit more from authenticity than others.
- Study 2 partially supported the causal prediction: authentic posting improved mood and positive affect and reduced negative affect within persons, but did not significantly change Life Satisfaction over the 1-week period.

b. Include Key Figures/Tables

- Table 1: Regression models showing positive coefficients for Quantified Authenticity predicting Life Satisfaction in both Likes-based and language-based models, with and without controls.

	Model 1			Likes Model 2			Model 3		
	B	SE(B)	β	B	SE(B)	β	B	SE(B)	β
QA	0.156***	0.014	0.113	0.069***	0.013	0.050	0.070***	0.013	0.051
Extr.	-	-	-	0.027*	0.013	0.020	0.022	0.014	0.016
O	-	-	-	-0.025*	0.013	-0.018	-0.016	0.013	-0.012
C	-	-	-	0.161***	0.013	0.117	0.165***	0.014	0.120
E	-	-	-	0.151***	0.014	0.109	0.155***	0.014	0.112
A	-	-	-	0.080***	0.014	0.058	0.081***	0.014	0.059
N	-	-	-	-0.511***	0.015	-0.371	-0.517***	0.015	-0.375
QA × O	-	-	-	-	-	-	0.026	0.014	0.018
QA × C	-	-	-	-	-	-	0.010	0.014	0.008
QA × E	-	-	-	-	-	-	0.014	0.014	0.011
QA × A	-	-	-	-	-	-	-0.002	0.013	-0.001
QA × N	-	-	-	-	-	-	-0.021	0.016	-0.015
Adj-R ²		0.01			0.26			0.26	

	Language			Model					
	B	SE(B)	β	B	SE(B)	β	B	SE(B)	β
QA	0.119***	0.025	0.085	0.046*	0.023	0.032	0.059*	0.024	0.042
Extr.	-	-	-	0.005	0.023	0.003	0.001	0.024	0.001
O	-	-	-	-0.029	0.022	-0.021	-0.019	0.023	-0.013
C	-	-	-	0.157***	0.023	0.111	0.159***	0.024	0.113
E	-	-	-	0.150***	0.024	0.106	0.153***	0.024	0.109
A	-	-	-	0.123***	0.023	0.087	0.119***	0.024	0.085
N	-	-	-	-0.503***	0.026	-0.357	-0.516***	0.026	-0.366
QA × O	-	-	-	-	-	-	0.027	0.021	0.022
QA × C	-	-	-	-	-	-	-0.005	0.023	-0.004
QA × E	-	-	-	-	-	-	0.022	0.023	0.016
QA × A	-	-	-	-	-	-	-0.015	0.024	-0.011
QA × N	-	-	-	-	-	-	-0.059	0.027	-0.045
Adj-R ²		0.01			0.25			0.25	

Models presented include Quantified Authenticity (QA-Euclidean Distance) on Life Satisfaction, with controls, and interaction effects of QA and the Big Five personality traits. (Likes-Based Model N = 9237; Language-Based Model N = 3215). ***p < 0.001, **p < 0.01, *p < 0.05.

- Figure 1: Robustness figure summarizing 16 model specifications; 11 of 16 estimates are statistically significant and all coefficients are in the positive direction.

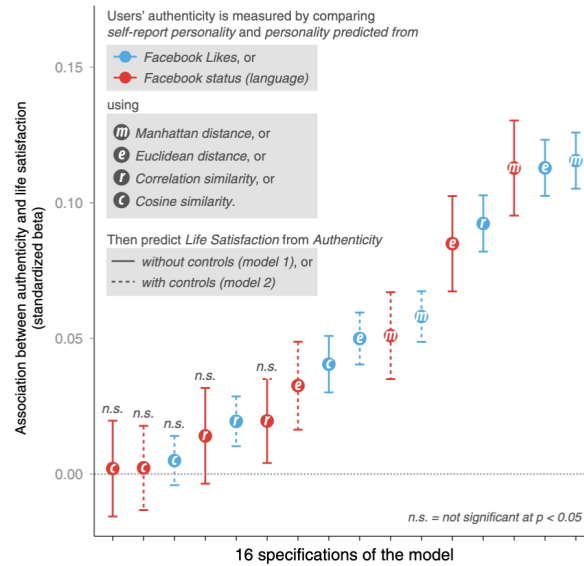


Fig. 1 Quantified authenticity on Life Satisfaction. Figure 1 presents standardized beta coefficients for Quantified Authenticity using ordinary least squares regressions in 16 individual regressions predicting Life Satisfaction. Quantified Authenticity is significantly associated with Life Satisfaction in 11 out of the 16 models. Quantified Authenticity is measured as the consistency between self-reported personality and two other sources of personality data: language and Likes, respectively, (indicated in red and blue color). Quantified Authenticity is defined using four distance metrics, respectively: Manhattan, Euclidean, correlation, and cosine similarity (indicated with a letter in the dots). Models with and without control variables are indicated with dashed and solid line, respectively.

- Figure 2: Study 2 experimental design showing the 2-week crossover structure.

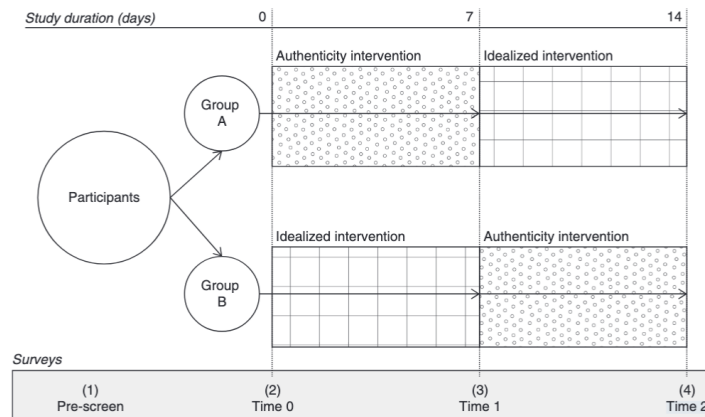


Fig. 2 Study 2 experimental design. Figure 2 presents the longitudinal experimental study design for Study 2 with key timepoints, interventions, and surveys.

- Figure 3: Standardized mean well-being outcomes by condition and time, showing clearer within-person benefits of authentic posting for mood and affect than for Life Satisfaction.

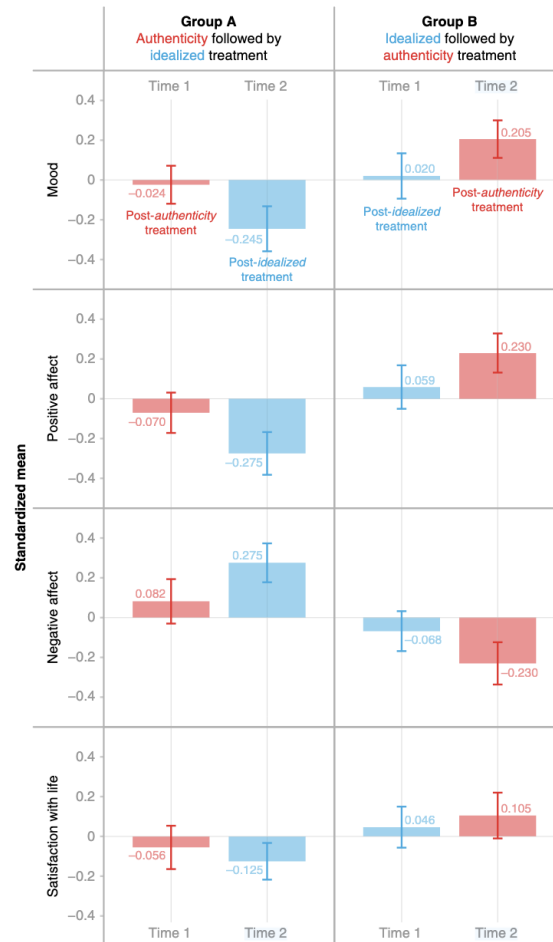


Fig. 3 Experimental outcomes by authenticity and idealization treatment. The bar charts illustrate the standardized mean of well-being indicators (mood, positive affect, negative affect, and Life Satisfaction) across two study time points by condition. The red bars indicate scores for the weeks in which participants were asked to post authentically, and the blue bars scores for the weeks in which they were asked to post in a self-idealized way. Error bars represent standard errors. The left-side panel presents Group A who received the authenticity treatment followed by the idealized treatment. The right-side panel presents Group B who received the idealized treatment followed by the authenticity treatment. This experiment was conducted once with independent samples in each group.

c. Authors' Conclusions

The authors conclude that authentic self-expression on social media is associated with better subjective well-being and that the effect depends less on whether people use social media than on how they use it. They argue that authentic posting appears beneficial across personality types and that short-term well-being improves when people post more authentically rather than more idealized.

4. Discussion

a. Implications

- The paper helps reconcile mixed findings on social media and well-being by shifting attention from amount of use to style of self-expression.
- It demonstrates the value of computational social science methods for measuring difficult constructs behaviorally rather than relying only on self-report.
- It suggests that authenticity may be a psychologically beneficial online practice, especially for short-term affect.

b. Limitations

- Study 1 is correlational, so causal direction cannot be established there.

- Study 2 has a small convenience sample and reduced statistical power because the preregistered sample size was not reached.
- The manipulation was more successful within persons than between persons, suggesting authentic vs. idealized posting is not easy to shift in a one-shot intervention.
- The work focuses on Facebook-era social media and on personality-based authenticity, so generalizability to other platforms and other forms of authenticity is uncertain.
- The study does not explain why people deviate from authenticity or whether doing so is intentional.

c. Future Directions

- Compare authentic social media use with no social media use at all.
- Test mechanisms that explain why authenticity improves well-being, such as self-verification, belonging, or reduced internal conflict.
- Examine platform differences and whether the effects replicate on newer social media environments.
- Investigate whether intentional versus unintentional self-discrepancy changes the strength of the well-being association.
- Use larger experimental samples and longer time frames to test effects on more stable outcomes such as Life Satisfaction.

Response Letter (~600 words)

• Your assessment of the study

Bailey et al. (2020) is a strong example of how computational methods can extend conventional communication research without losing sight of classic psychological questions. The article asks a simple but consequential question: when people use social media, is it better for well-being to present an idealized self or a more authentic one? What makes the paper especially compelling is that the authors do not rely only on self-reported authenticity. Instead, they introduce “Quantified Authenticity,” a behavioral measure based on the distance between self-reported personality and personality predicted from Facebook Likes and status updates. This methodological move addresses a key limitation in prior research, where self-reports of authenticity may already be shaped by mood, self-image, or social desirability.

In Study 1, the large sample size and the use of digital trace data provide strong external validity. The authors show that individuals whose online self-expression more closely matches their self-view report greater Life Satisfaction, and this relationship remains robust across different operationalizations of authenticity and after controlling for personality. Importantly, the findings do not support an alternative explanation based on socially desirable personality profiles, suggesting that the benefits of authenticity are not limited to certain personality types.

Study 2 further strengthens the paper by addressing the limitation of causal inference. The preregistered crossover design is particularly effective, as each participant experiences both authentic and self-idealized posting conditions, allowing for within-person comparisons. The results are appropriately nuanced: authentic posting improves mood and affect but does not significantly change Life Satisfaction over a short time frame. This distinction aligns with theory, as affective outcomes are more immediate, whereas Life Satisfaction is a more stable cognitive evaluation.

• Weaknesses

The main limitations of the study relate to scope rather than execution. The observational data are drawn from Facebook users in the myPersonality project, which may limit generalizability to other platforms or user populations. Similarly, the experimental sample is relatively small and not fully representative, which may reduce statistical power, especially for detecting effects on more stable outcomes such as Life Satisfaction.

In addition, the study operationalizes authenticity specifically as consistency between self-reported and behaviorally inferred personality. While this is a clear and innovative measure, it captures only one dimension of authenticity. Other aspects, such as emotional disclosure or situational self-presentation, are not directly addressed. Finally, although the study demonstrates a relationship between authenticity and well-being, it does not directly test mediating mechanisms, leaving the underlying psychological processes somewhat inferential.

• Other notes

Overall, the study makes a valuable contribution by reframing the effects of social media use. Rather than treating social media as inherently beneficial or harmful, the authors show that outcomes depend on how individuals engage with these platforms. The integration of computational measures with experimental design is particularly noteworthy, as it combines the strengths of large-scale behavioral data with causal inference.

The figures also play an important role in supporting the analysis. Figure 1 effectively demonstrates robustness across multiple model specifications, while Figures 2 and 3 clearly

illustrate the experimental design and the distinction between short-term affective outcomes and longer-term life satisfaction. More broadly, the article serves as a strong example of how computational approaches can complement, rather than replace, traditional psychological theory and experimental methods.